

Exploratory Data Analysis

Data Preprocessing & Basic Data Visualization using R

2026. 9. 16.



What EDA is for

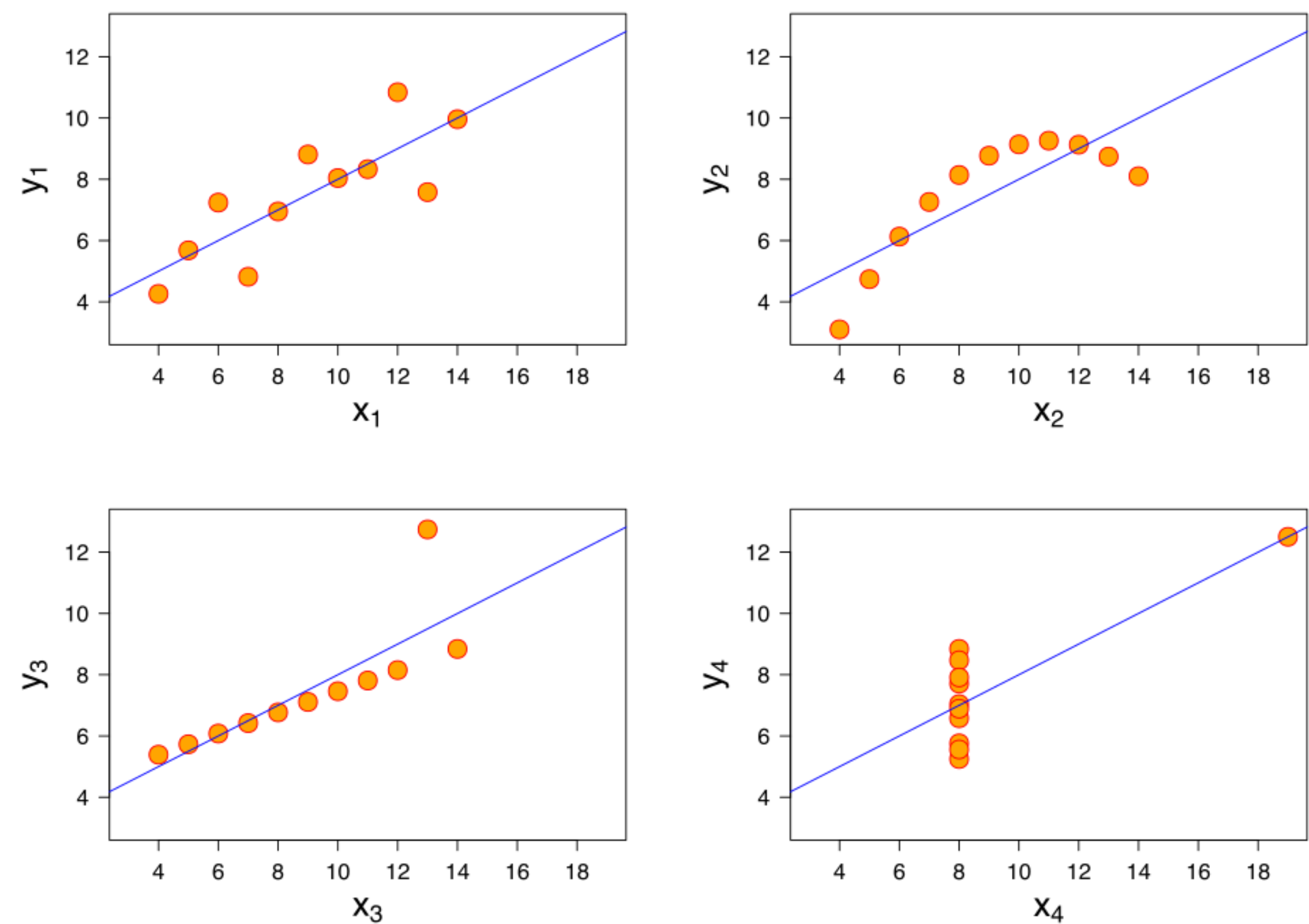


“The greatest value of a picture is
when it **forces us to notice**
what we never expected to see.”

John W. Tukey

Why We Visualize Data Before Drawing Conclusions

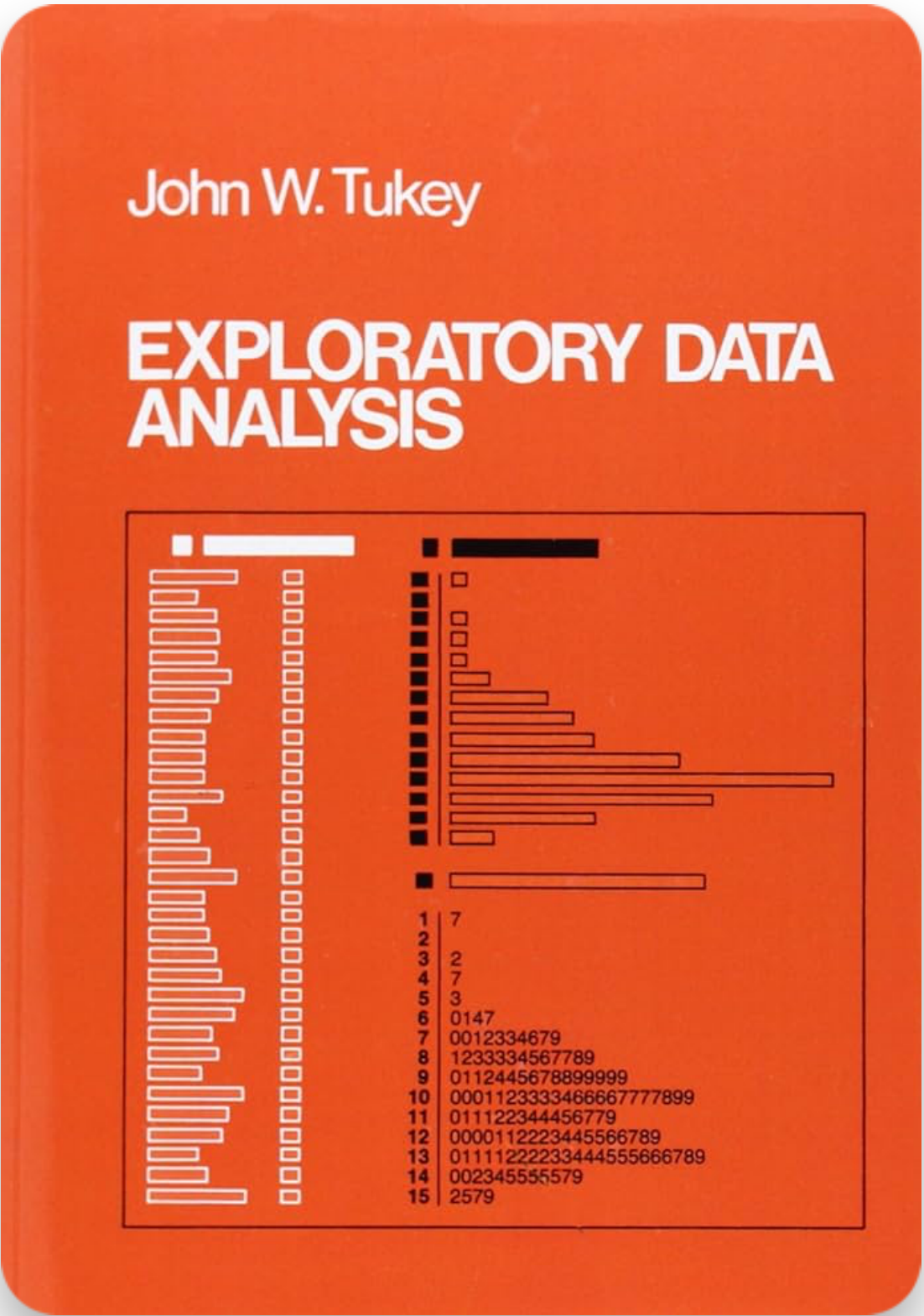
- Francis Anscombe (1973): “numerical calculations are exact, but graphs are rough”



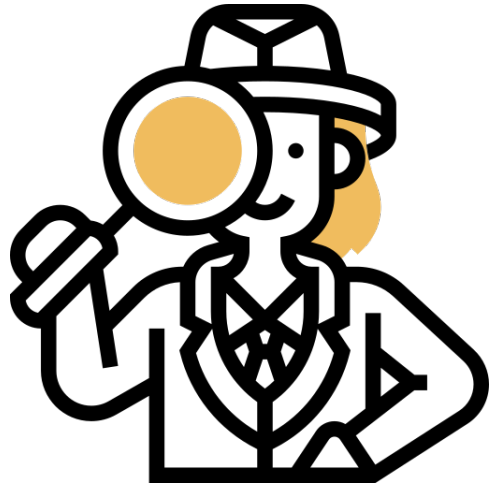
For all four datasets:

Property	Value	Accuracy
Mean of x	9	exact
Sample variance of x: s_x^2	11	exact
Mean of y	7.50	to 2 decimal places
Sample variance of y: s_y^2	4.125	± 0.003
Correlation between x and y	0.816	to 3 decimal places
Linear regression line	$y = 3.00 + 0.500x$	to 2 and 3 decimal places, respectively
Coefficient of determination of the linear regression: R^2	0.67	to 2 decimal places

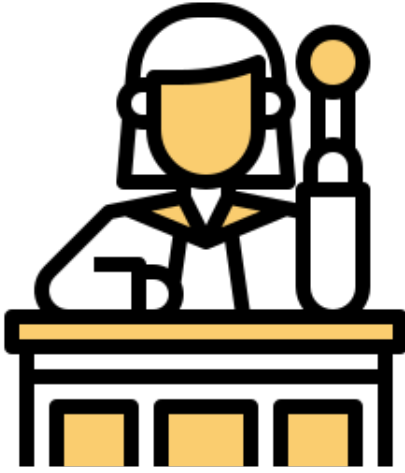
Introduction to EDA



- It is important to understand what you CAN DO before you learn to measure how WELL you seem to have DONE it.

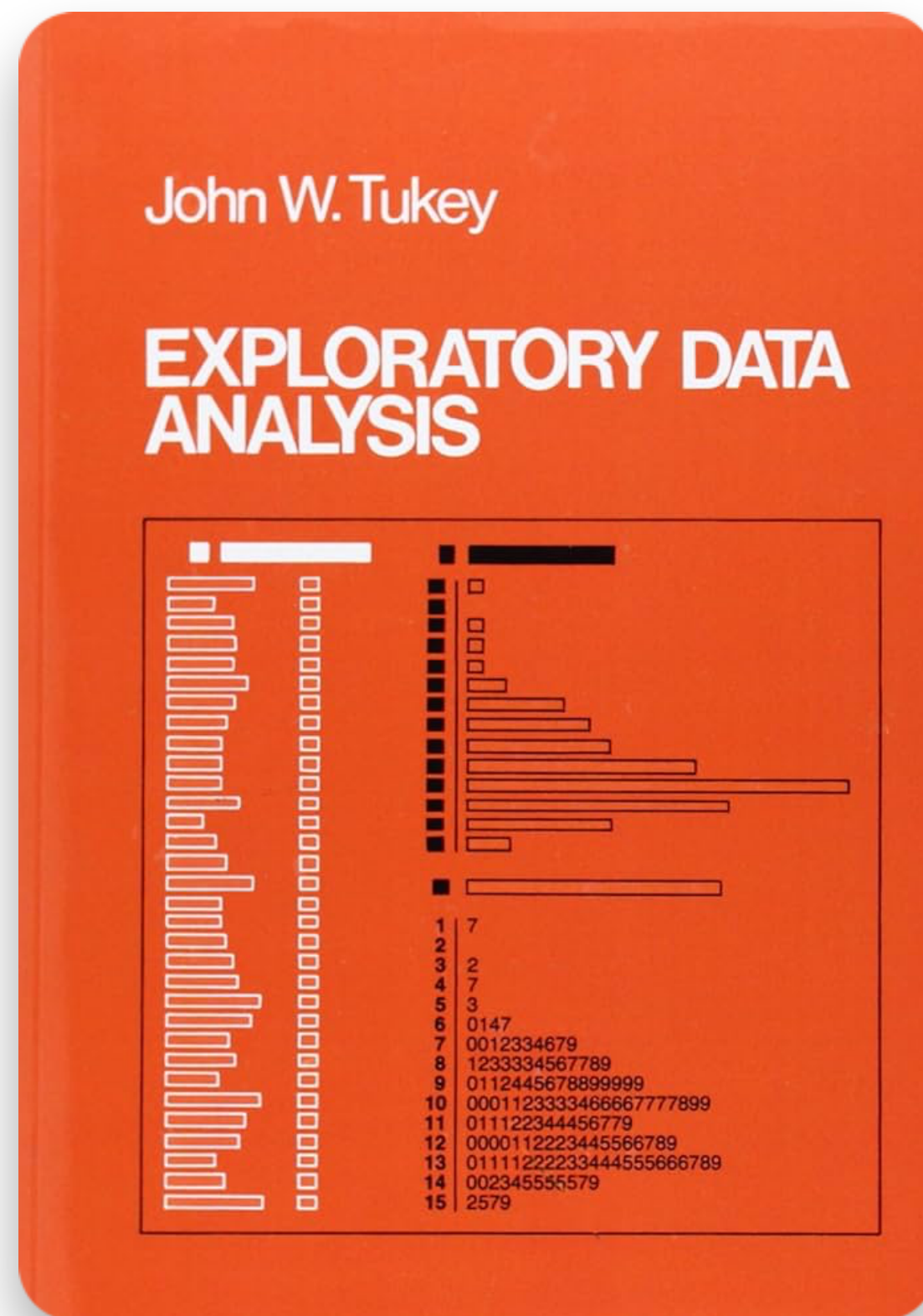


Detective
: search for the evidence

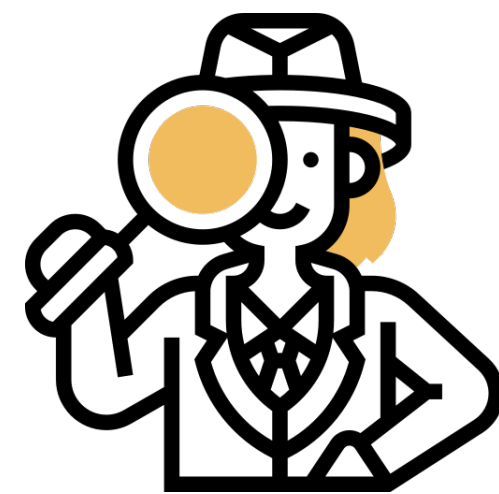


Judge
: evaluation of the evidence's strength

Introduction to EDA



- Exploratory data analysis can never be the whole story, but nothing else can serve as the foundation stone--as the first step.



Exploratory
data analysis



Confirmatory
data analysis

Common Pitfalls in EDA

Data $\neq$ Reality

- System artifacts $\neq$ real behavior
- Meaning depends on context

Missing $\neq$ Zero

- Missing values carry information
- Treating them as zero biases results

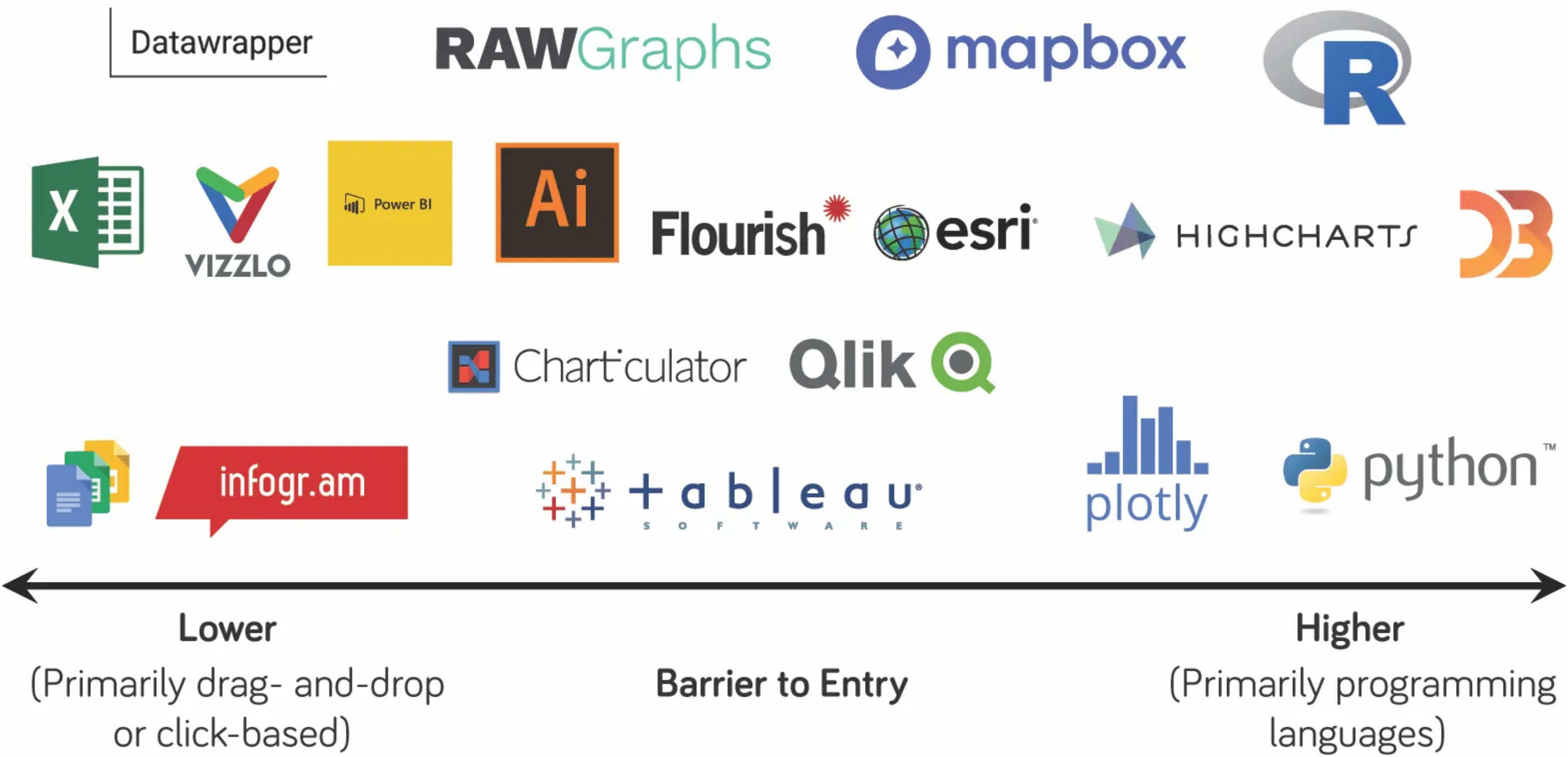
Patterns $\neq$ Evidence

- Noise can look meaningful
- Correlation $\neq$ causation

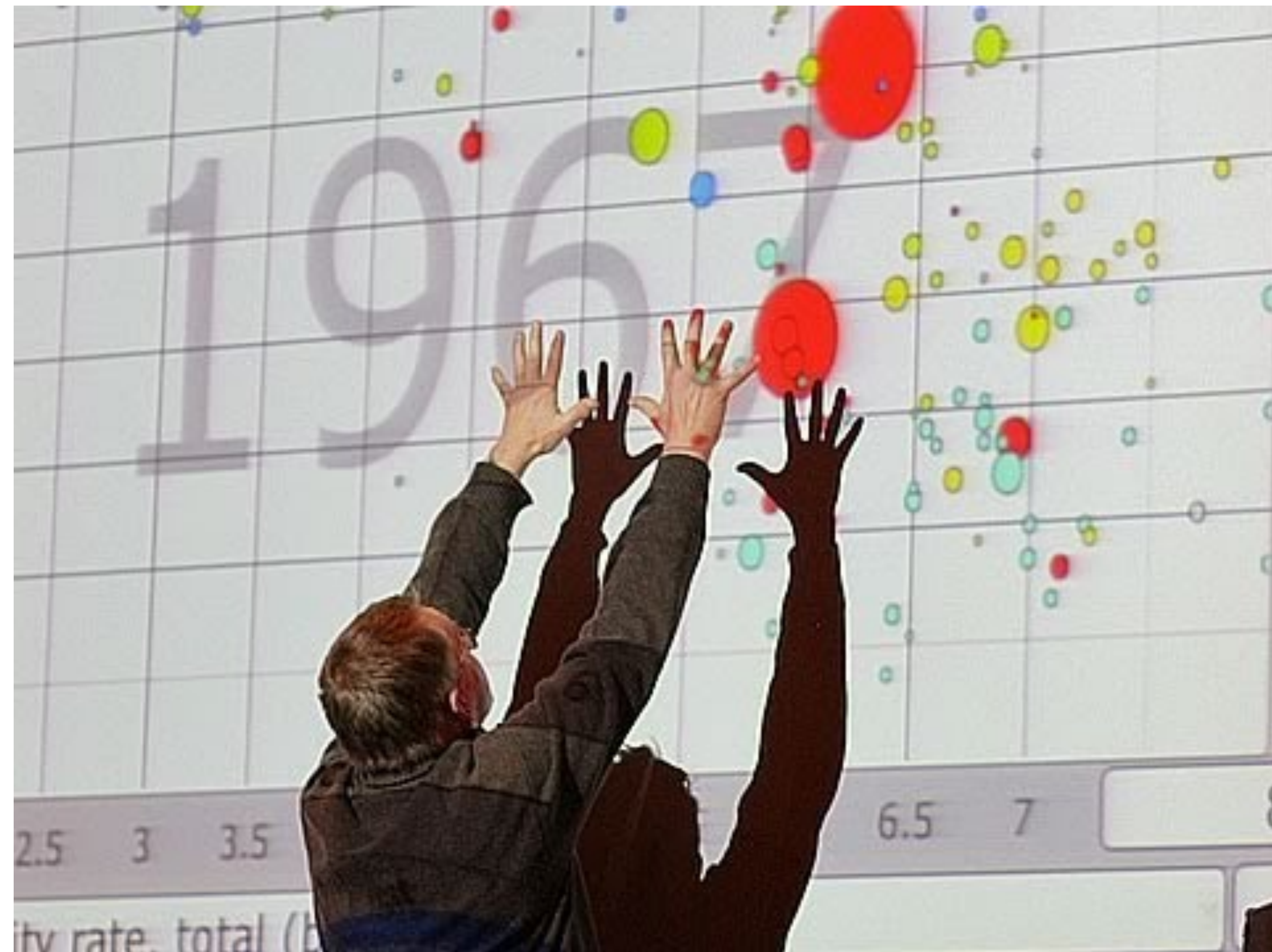
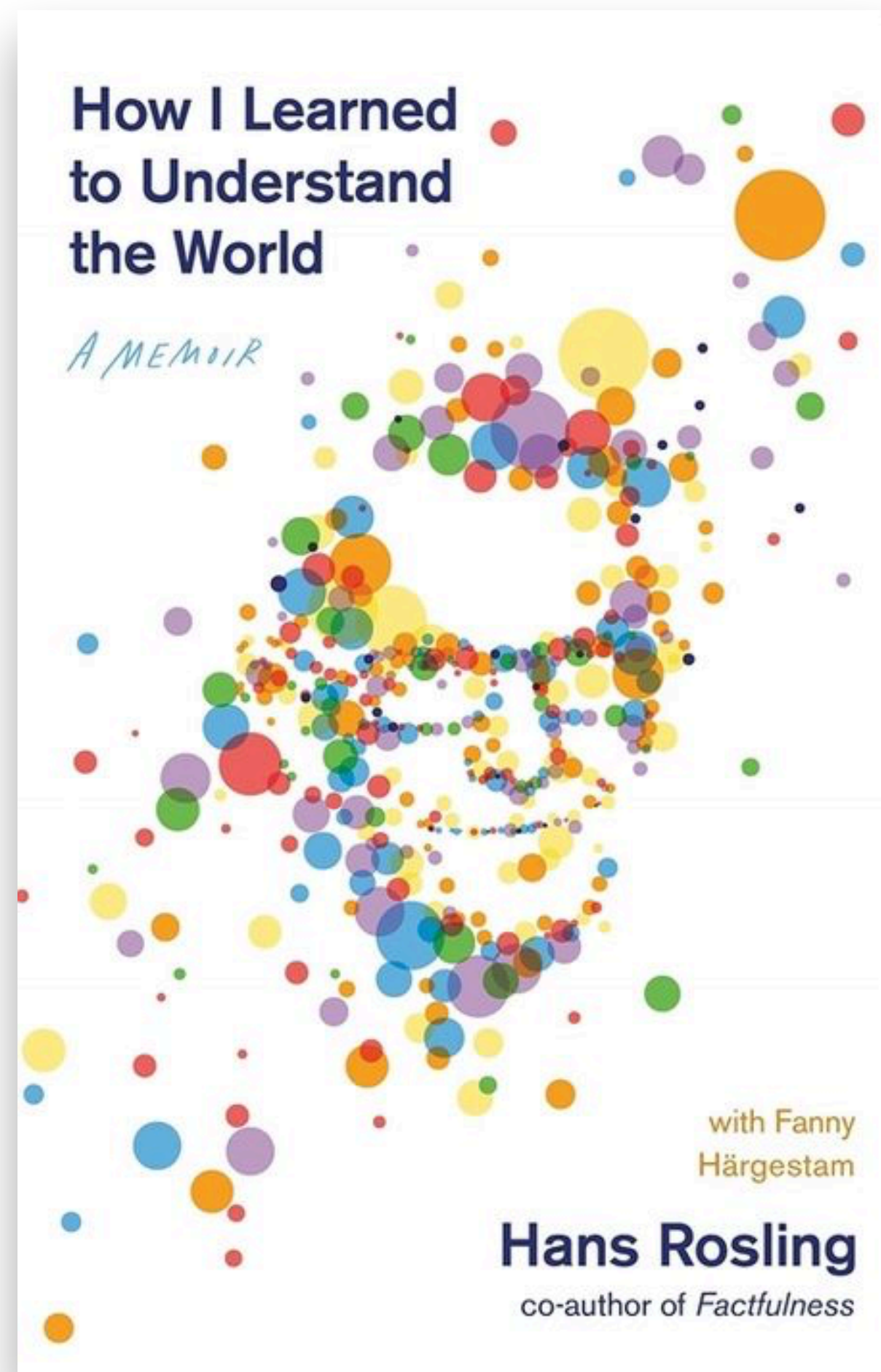
Aggregates Hide Structure

- Segmentation matters
- Different groups, different patterns

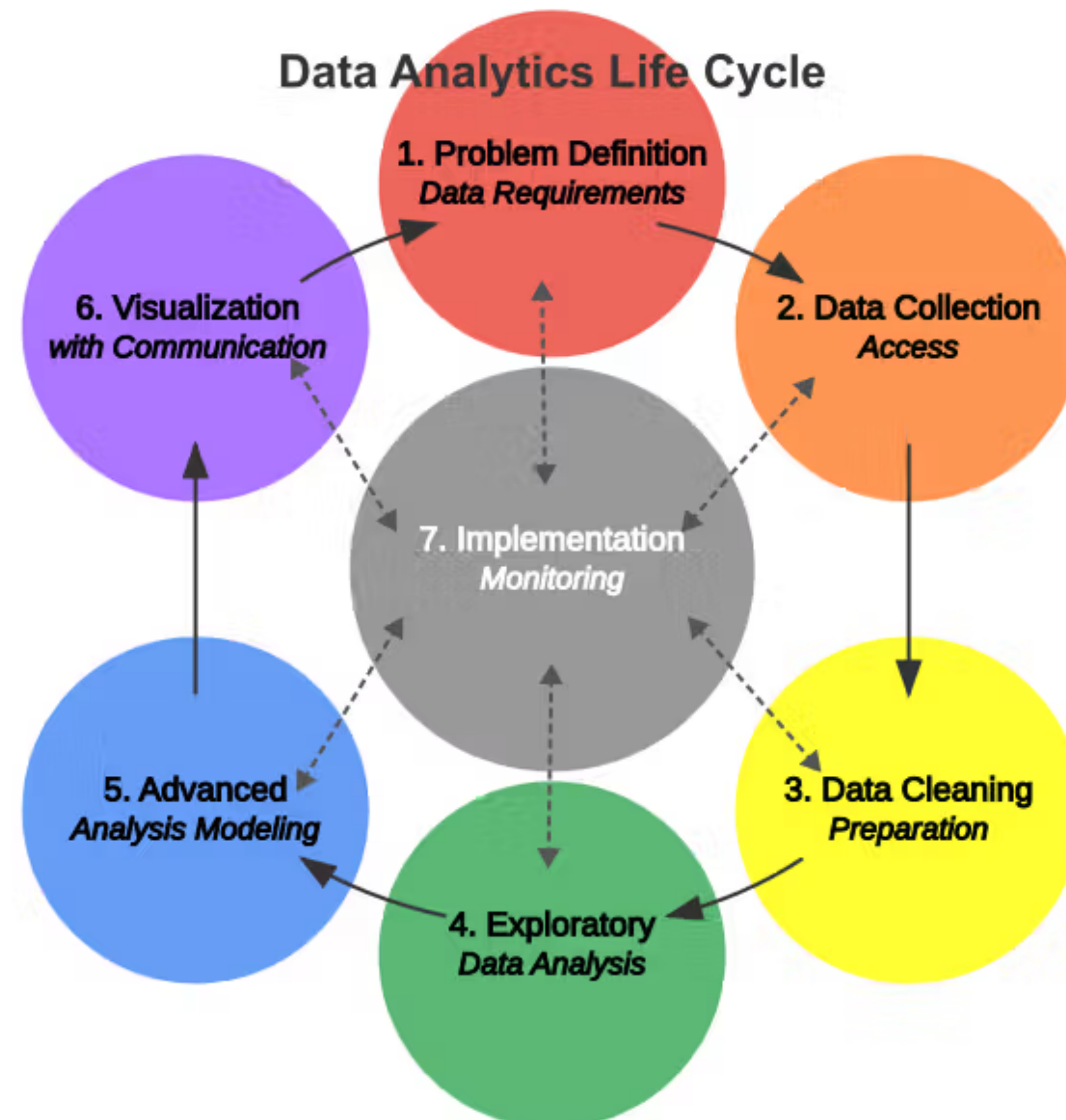
EDA with Visualization



The Best Stats You've Ever Seen



An EDA Lifecycle



Legend: The hub-and-spoke model of data analytics where Implementation Monitoring connects with all phases of the analytical process

Phase 4: Exploratory data analysis (EDA)

Once data is cleaned and prepared, exploratory data analysis (EDA) provides the first opportunity to understand what the data reveals. This critical phase helps analysts:

- Understand the distribution and characteristics of key variables
- Identify relationships and correlations between variables
- Discover patterns, trends, and anomalies
- Generate initial hypotheses for deeper investigation
- Validate assumptions about the data

Keep in Mind

“... **many problems do not have a single right answer.** There can be many ways to approach a body of data. ... Accordingly, it will often be quite reasonable for different analysts to reach somewhat different analyses.”

John W. Tukey